

Qatar-1b

R-2709

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130915 · created 2026-09-01T01:00:40+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456550.76852 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.036
Transit depth (Rp/Rs)^2	3.63 +/- 1.35 [%]
Orbital Inclination (inc)	82.89 +/- 2.23
Airmass coefficient 1 (a1)	1.132 +/- 0.049
Airmass coefficient 2 (a2)	-0.081 +/- 0.058
Scatter in the residuals of the lightcurve fit is	4.35 %
Best Comparison Star	#1 - [373, 289]
Optimal Aperture	2.29
Optimal Annulus	5.5
Transit Duration (day)	0.062 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.036 vs 0.1463 (deviation 1.21σ)
Duration fit vs published	0.062 d vs 0.0692 d (deviation 0.38σ)
Mid-time O–C	-0.1 min
Depth SNR	5.3
Residual scatter	4.35 %

Light curve

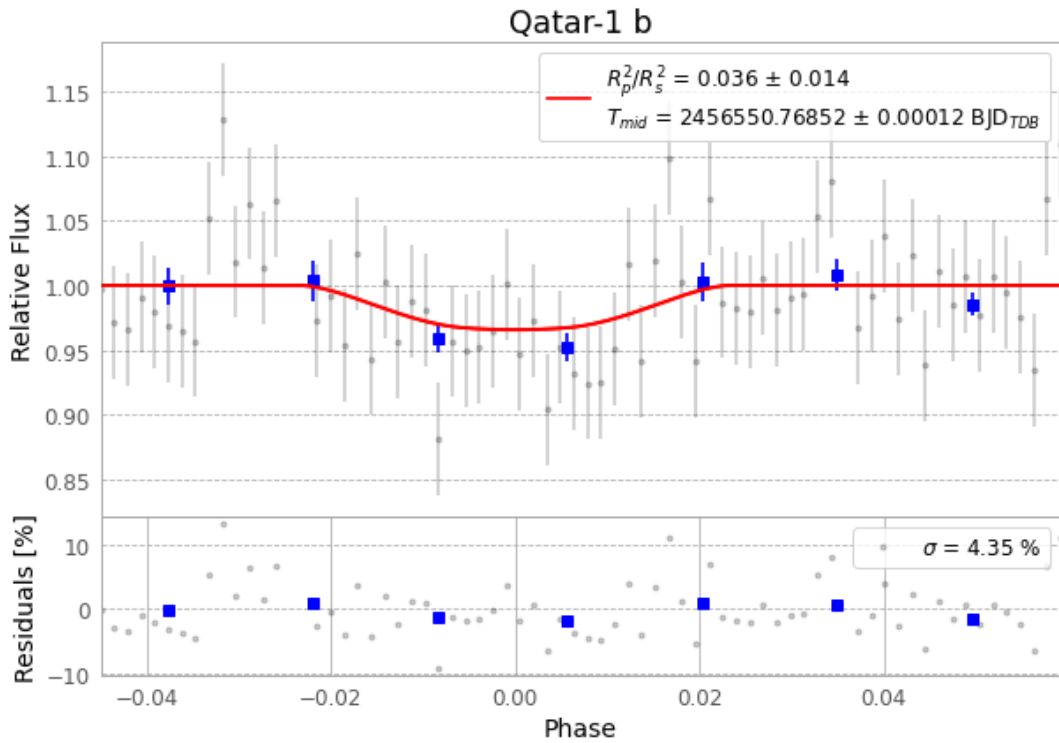


Figure 1 — FinalLightCurve_Qatar-1 b_2013-09-14.png (SHA-256 655f6111e5c39b62...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [442, 266], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 143a032a1b265c9a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1130915045112.FITS → Qatar-1130915082412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130915.log (650ac24b6a9f...), FinalLightCurve_Qatar-1 b_2013-09-14.png (655f6111e5c3...), FinalLightCurve_Qatar-1 b_2013-09-14.csv (b51e6f5b9752...)

Compute resources

Wall time 112.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:00Z.